



SINOP LANDFILL GAS TO ELECTRICITY PROJECT



Project title	Sinop Landfill Gas to Electricity Project
Project ID	VCS 4014
Monitoring period	01-March-2023 to 31-August-2024
Original date of issue	20-August-2024
Most recent date of issue	05-June-2025
Version	05
VCS Standard Version	VCS 4.6
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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Sinop Landfill Gas to Electricity Project (hereafter referred to as the project) is located at Sinop Landfill Site in Hacıoglu Neighborhood, Central District, Sinop Province Turkey. The project is owned by Biokütle Enerji Elektrik Üretim A.Ş. (hereafter referred to as the project owner). The project is designed to utilize the gas produced in the existing landfill to generate electricity. The landfill site has been in this area since 2012. The provisional acceptance of the facility is 29-June-2021¹ with half capacity of 1.453 MWm / 1.414 MWe with single generator, but it started its operation as of 01-July-2021. The total installed capacity will be 2.906 MWm / 2.828 MWe (2x1.453) MWm / (2x1.414) MWe, consisting of 2 sets of generators, and installation of second generator is planned to be completed as of 2025. As per the generation licence, the feed-in electricity is estimated to be 10,605 MWh/year with half capacity and 21,210 MWh/year¹ with full capacity.

The wastes coming to the field come from the member districts of the Sinop Coastal Municipalities Union. The member districts of the union are; Sinop center, Gerze, Dikmen, Erfelek, Ayancık and Türkeli. As per the weighbridge record, 78,731.11 tons of MSW² was handled during the monitoring period.

Prior to the implementation of the project, the landfill site did not have gas extraction and utilization systems installed, and the LPG was directly emitted to the atmosphere. The project collects and utilizes LFG generated by Sinop landfill for power generation. Therefore, the project drastically reduces methane emissions, and at the same time, electricity is generated from the collected gas. Electricity generated is provided to the grid and the facility utilizes electricity generated by the project instead of drawing electricity from the grid. The environmental, socio-economic, and technological benefits of the Project are described as follows:

- Reduction in fossil fuel use (imported or local) by using renewable energy resources,
- Reduction in greenhouse gas emissions from the landfill area by using biogas for electricity production,
- Reduction in energy production costs and imported energy amounts,
- Improvement of environmental conditions (GHG and odor) and safety in the landfill area.

The project has reduced 43,752 tCO₂e/year and generated net 8,788.376 MWh electricity in total date of 01-March-2023 and ending date of 31-August-2024.

¹ Generation licence

² Weighbridge records, weighbridge is calibrated and the latest calibration dates are: 04-August-2022, 28-October-2024

¹ Sinop Landfill Provisional Acceptance

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Joint Validation & Verification	01-July-2021 to 28-February-2023	VCS	Applus	One year and 8 months
Verification	01-March-2023 to 31-August-2024	VCS	Bureau Veritas	One year and 6 months

1.3 Sectoral Scope and Project Type

Sectoral scope	Scope 1
Project activity type	Energy industries (renewable-/non-renewable sources)
Sectoral scope	Scope 13
Project activity type	Waste handling and disposal

1.4 Project Proponent

Organization name	<i>Biokütle Enerji Elektrik Üretim Anonim Şirketi</i>
Contact person	<i>Savaş Uzun</i>
Title	<i>CEO</i>
Address	<i>Soğuksu Mah. 308 Sok. Adabel Sit. No:42/5 Muratpaşa, Antalya</i>
Telephone	<i>+90 (242) 237 31 00</i>
Email	<i>savas.uzun@biosenerji.com</i>

1.5 Other Entities Involved in the Project

Organization name	<i>Gaia Climate Finansal Danışmanlık Hizmetleri ve Ticaret A.Ş.</i>
Role in the project	<i>Carbon Certification Consultant</i>
Contact person	<i>Gediz S. Kaya</i>
Title	<i>Managing Partner</i>

Address	Maslak Mah. Maslak Meydan Sk. Beybi Giz Plaza A Blok No:1/99 Sarıyer/İstanbul
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Email	gkaya@gaiacclimate.com

1.6 Project Start Date

Project start date	01-July-2021
Justification	Project started to utilize methane and generate electricity as of 01-July-2021 which can be seen in EPIAS records. Therefore, start date of the project is 01-July-2021 as per VCS standard 4.4 paragraph 3.8.

1.7 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	01-July-2021 to 30-June-2028

1.8 Project Location

The project is in Hacıoglu Neighborhood, which is located in Merkez District of Sinop Province in Turkey. The GPS coordinates of the Project area are provided in the following table.

Table 1 Project Area Coordinates

N	E
41.949269823	35.036668530
41.949272309	35.036911608
41.949194300	35.036913527
41.949194861	35.036979850
41.948934092	35.036983778

41.948930761	35.036673646
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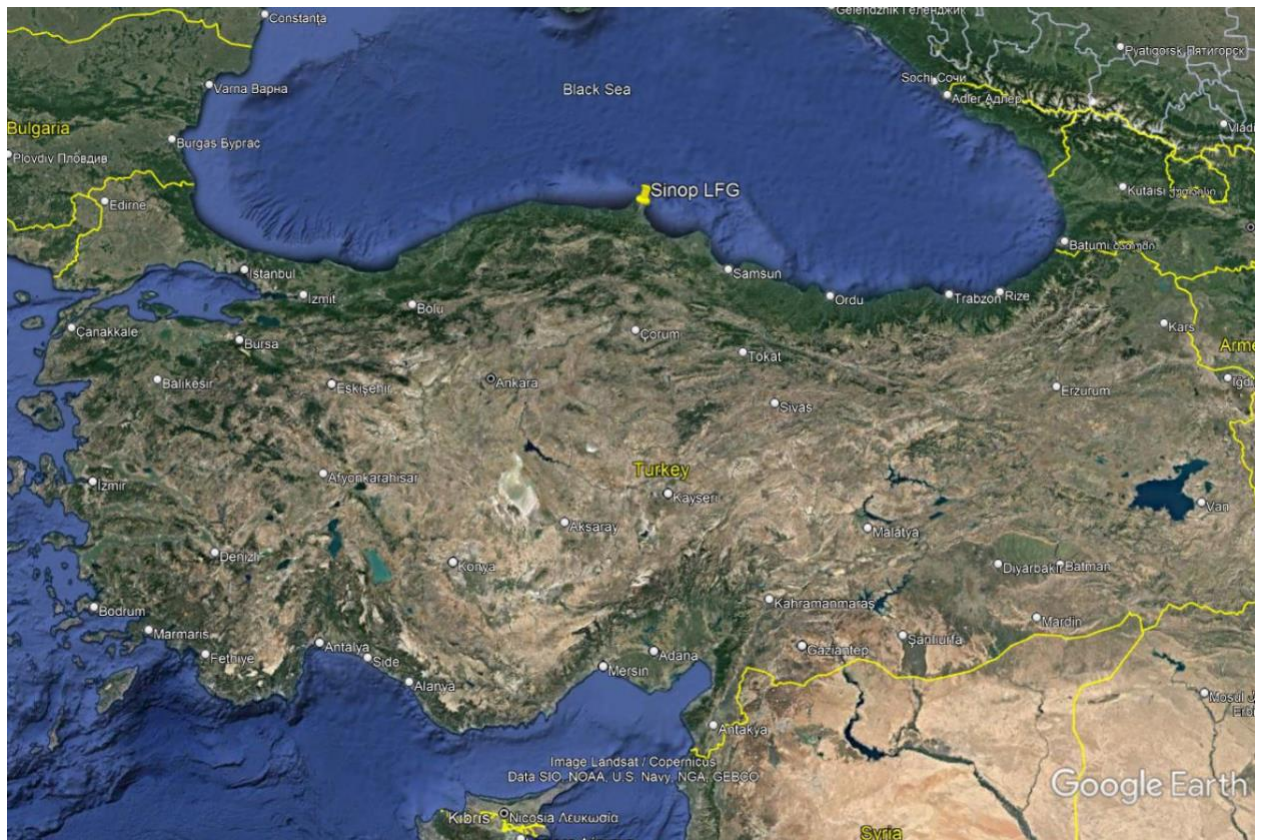


Figure 1 Project Location



Figure 2 Google Earth view of the project site

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.G	Landfill methane recovery	10.0
Tool	03	Tool to calculate project or leakage CO2 emissions from fossil fuel combustion	03 .0
Tool	04	Emissions from solid waste disposal sites	08.1
Tool	05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	09	Determining the baseline efficiency of thermal or electric energy generation systems	03.0
Tool	32	Positive lists of technologies	04.0

Methodology	AMS-I.D	Grid-connected renewable electricity generation	18.0
Tool	03	Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion	03.0
Tool	07	Tool to calculate the emission factor for an electricity system	07.0
Tool	10	Tool to determine the remaining lifetime of equipment	01
Tool	11	Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period	03.0.1
Tool	16	Project and leakage emissions from biomass	05.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.12 Sustainable Development Contributions

The project not only aims to generate energy from a sustainable and renewable energy source but also to find a sustainable solution to an unmanaged landfill which released landfill gas to atmosphere prior to the installation of facility. The Project will contribute to long term sustainable development of the region.

Main contributions to the Sustainable Development Goals¹⁷ adopted by the United Nations are summarized.

SDG 7: Affordable and Clean Energy

- The project converted previously unmanaged landfill into a biogas-to-energy facility which is also a clean and efficient facility for waste disposal and handling.
- Generates energy from a renewable source.
- Provides energy to the grid from renewable a source and replaces energy generated by fossil fuel utilization.

SDG 8: Decent Work and Economic Growth

- Creates employment opportunities related to the management, operation and maintenance of landfill and the methane recovery and electricity generation systems.
- Turkey is an energy and fuel dependent country, and the project contributes to reduction of energy dependency for the country and thus contributes to economic growth.

SDG 13: Climate Action

- Management of unmanaged landfill site by establishing a new, clean and efficient facility for waste disposal and handling.
- Collection of landfill gas, reduction and prevention of direct emission of GHG to the atmosphere. The project achieved 43,752 tCO₂e/year emissions reductions between 01-March-2023 and 31-August-2024.

Table 2: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	Implemented activities to increase	The Project Activity is provided employment to 12 people.	Project Proponent provides employment to 12 people during the monitoring period complying with Labor Law in Turkey.
2)	8.8	Protect labour rights and promote safe working environments	Implemented activities to increase	During the monitoring period, a total of 4 trainings were provided to employees, 2 on environmental topics and 2 on occupational health and safety.	Occupational Health and Safety training is provided to employees at least once a year previous monitoring period. From 01-July 2021 onwards at least 5 trainings were provided to employees.
3)	7.2	By 2030, increase substantially the share of renewable energy in the global energy mix	Implemented activities to increase	Net electricity generation during the monitoring period is 8,788.376 MWh.	From 01-July-2021 onwards, 16,911.38 MWh of electricity from renewable sources has been supplied to the power grid.

4)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	Total emission reduction during the monitoring period is 43,752 tCO ₂ e.	From 01-July-2021 onwards, total emission reduction of the project is 84,884 tCO ₂ e.
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1.13 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	To identify stakeholders likely to be impacted by the project, the Project Proponent (PP) assessed the surrounding area and identified nearby settlements, Hacıoğlu neighborhood, as potentially impacted communities due to their proximity to the Sinop landfill site (please note that the landfill site is located at the Hacıoğlu neighborhood). In addition, potential stakeholders, including NGOs, ministries, government were identified. Through this process, the following stakeholders were identified: local residents of Hacıoğlu neighborhood, the surrounding settlements, municipality, the Muhtar, ministries, NGOs, and project owner. The local stakeholders were informed about the stakeholder meeting via announcements which were posted on 05-October-2022. The announcements were placed around high populated areas of the neighborhood to be seen by the locals. The potential stakeholders were invited to the meeting via official emails sent on 6 October 2022.
Legal or customary tenure/access rights	N/A as the project is implemented in the landfill site and project owner has the rights of carbon project.
Stakeholder diversity and changes over time	The project has no impact on the social, economic, and cultural diversity within stakeholder groups

Expected changes in well-being	The project is expected to improve local environment, provide work opportunities, rehabilitate (forest the exhausted landfill lots and supply clean energy
Location of stakeholders	Nearby settlements, Sinop municipality.
Location of resources	N/A

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	Most common method used by local shareholders are communicating with Plant Manager via Most common method used by local shareholders are communicating with Plant Manager via village representative (Mukhtar). Plant managers are directly responsible for keeping communication with local s directly or through Mukhtars
Date(s) of stakeholder consultation	18-October-2022
Communication of monitored results	There is no complaints on the grievance logbook during monitoring period.
Consultation records	N/A
Stakeholder input	During the meeting a presentation was provided detailing the project, its grievance mechanisms, environmental and social impacts, contributions to sustainable development goals. In addition, during the local stakeholder consultation meeting, participants filled in an evaluation form regarding the project to document their feedback on the impacts of the project. In general, local stakeholders were supportive of the project activity. Most of the local stakeholders thought the Project would help improve the livelihood of local people and promote local economic development without much adverse environmental impact. Therefore, the project implementation is regarded as beneficial by most of the stakeholders. During the meeting, the stakeholders did not raise any concerns about the project, and they indicated since the implementation of the project waste and odor issue caused by the wild dumpsites disappeared.

2.1.3 Free, Prior, and Informed Consent

Consent	The project does not affect Indigenous Peoples, customary rights holders, or traditional lands. Therefore, this section is not applicable. However, the project proponent conducted a local stakeholder consultation on 18 October 2022, informing and consulting local communities, particularly the residents of Hacıoğlu Neighborhood where the landfill is located. Announcements were made in public areas, and formal invitations were sent to relevant local authorities and stakeholders. No objections were raised during the consultation, and there were no pre-existing or ongoing conflicts in the area. All stakeholders had access to project information prior to the consultation and had the opportunity to provide feedback.
Outcome of FPIC	The project has not resulted in any encroachment on land, nor has it caused any physical or economic displacement. There is an official “EIA Not Required” letter obtained from the Sinop Governorship Provincial Directorate of Environment and Urbanization, which clearly states the addresses and provides a description of the project activities. This confirms that the project is implemented within a legally designated area and has received the necessary administrative approvals, ensuring that there is no infringement on community or customary rights. No unresolved conflicts or opposition were identified during the stakeholder consultation process.

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
N/A	The project owner didn't receive any grievance during the monitoring period. The grievance mechanism was established for stakeholders to forward their requests or concerns regarding the project activities at the Mukhtar. There is a logbook for the grievance mechanism at the Mukhtar. The

Mukhtar has all the contact information of the responsible persons from the project activity. Whenever a complaint arises from the local people or Mukhtar himself arises a concern regarding the project activity, he can directly contact the relevant responsible persons.

2.1.5 Public Comments

Summary of comments received	Actions taken
N/A	Following initial project listing, the VCS/Verra project Sinop Landfill Gas to Electricity Project was open for 30 days of public consultation from 12-January 2023 to 11-February-2023. No comment has been received during public consultation

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Biokütle Enerji Elektrik Üretim Anonim Şirketi has several landfill plants in Türkiye.

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risks are identified.	Although no specific risks to stakeholders' wellbeing have been identified, the project is implemented in full compliance with all relevant Turkish laws and regulations, which serve as preventative measures.
Risks to stakeholder participation	No risks are identified.	The project owner encourages continuous ongoing feedback from its stakeholders therefore the risk in this content is not identified to this project.
Working conditions	Working conditions are determined by Turkish labor law	Occupational Health and Safety specialist is follow the conditions.

	numbered 4857 . No additional risks are identified. ³	
Safety of women and girls	No risks are identified.	The women and girls are treated as per the law of land with no discriminations on gender basis.
Safety of minority and marginalized groups, including children	No risks are identified.	The project has no impact to safety of minority and marginalized groups, including children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risks are identified.	Environmental consultant follows the situation of the project.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified	Mitigation or preventative measure(s) taken
Discrimination	No risk identified.	The project operates in compliance with Turkish laws and regulations. As stated in Article 5 of the Labor Law No. 4857 ⁴ , the employer is obliged to treat all employees equally.
Sexual harassment	No risk identified.	The project operates in compliance with Turkish laws and regulations. Sexual harassment is prohibited and considered a criminal offence under Turkish law.

³ <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4857.pdf>

⁴ <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=4857&MevzuatTur=1&MevzuatTertip=5>

Gender equity in labor and work	No risk identified.	The management approach in other PP facilities is the same, and no discrimination is practiced. All facilities comply with Turkish labor law. This is a legal requirement. The PP has been required by parent company and local government to ensure gender equality in labor and work.
Forced labor	No risk identified.	In Türkiye force is an offence by Turkish punishment law (Item 79/1, last article).
Child labor	No risk identified.	In Türkiye child labor is an offence by Turkish punishment law .
Human trafficking	No risk identified.	In Türkiye human trafficking is an offence by Turkish punishment law.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	The project operates in Türkiye, where there are no groups legally or customarily recognized as Indigenous Peoples under the definitions of international human rights law, the UN Declaration on the Rights of Indigenous Peoples, or ILO Convention 169.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified.	The project operates in Türkiye, where there are no groups legally or customarily recognized as Indigenous Peoples under the definitions of international human rights law, the UN Declaration on the Rights of Indigenous Peoples, or ILO Convention 169.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	N/A

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	The project does not impact property rights.
Benefit sharing during the monitoring period	The project does not impact property rights.

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure(s) taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified	The project activity is exempt from EIA by official decision and according to this decision the project does not impact biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The project activity has no impact in this regard
Water consumption and stress	No risk identified	Under project activity, no direct process consumes freshwater, and wastewater is sent to the municipality wastewater treatment plant via sewage truck.

2.4.1 Rare, Threatened, and Endangered species

Species or habitat	This project is not located in, or adjacent to habitats for rare, threatened, or endangered species. Therefore, this section is not applicable.
Areas needed for habitat connectivity	This project is not located in areas needed for habitat connectivity. Therefore, this section is not applicable.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	This project is not located in, or adjacent to habitats for rare, threatened, or endangered species.
Areas for habitat connectivity	No risk identified	This project is not located in areas needed for habitat connectivity.

2.4.2 Introduction of species

N/A

2.4.3 Ecosystem conversion

N/A

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The Project, which is located in the Merkez District of Sinop, utilizes the landfill gas (LFG) generated by the existing landfill for energy generation. The Project has an operation period of 20 years under the current Project Owner. The Project Owner will then transfer the rights of operation of the Project to the Sinop Municipality at the end of the period.

Waste projections conducted under feasibility studies indicate that the landfill receives 173.6 t/day waste on average and the waste quantity is expected to rise during operation period from 155 t/day in 2021 to 236 t/day in 2040. As per the weighbridge record, 78,731.11 tons of MSW was handled during the monitoring period.

Waste is collected and deposited to the landfill by the Municipality. Then, landfill gas is extracted via landfill gas collection system. Collected gas is flared and utilized for electricity generation.

Steps in the landfill gas to electricity project:

1. Storage of waste in landfill
2. Generation of LFG at the landfill
3. Collection of LFG via pipelines
4. Booster system that draws collected LFG
5. Treatment of LFG via Chiller
6. Electricity and thermal energy generation in gas engines

7. Electricity feeding to national power grid
8. Retaining extra gas, that is not burnt in the engines, in the Gas holder

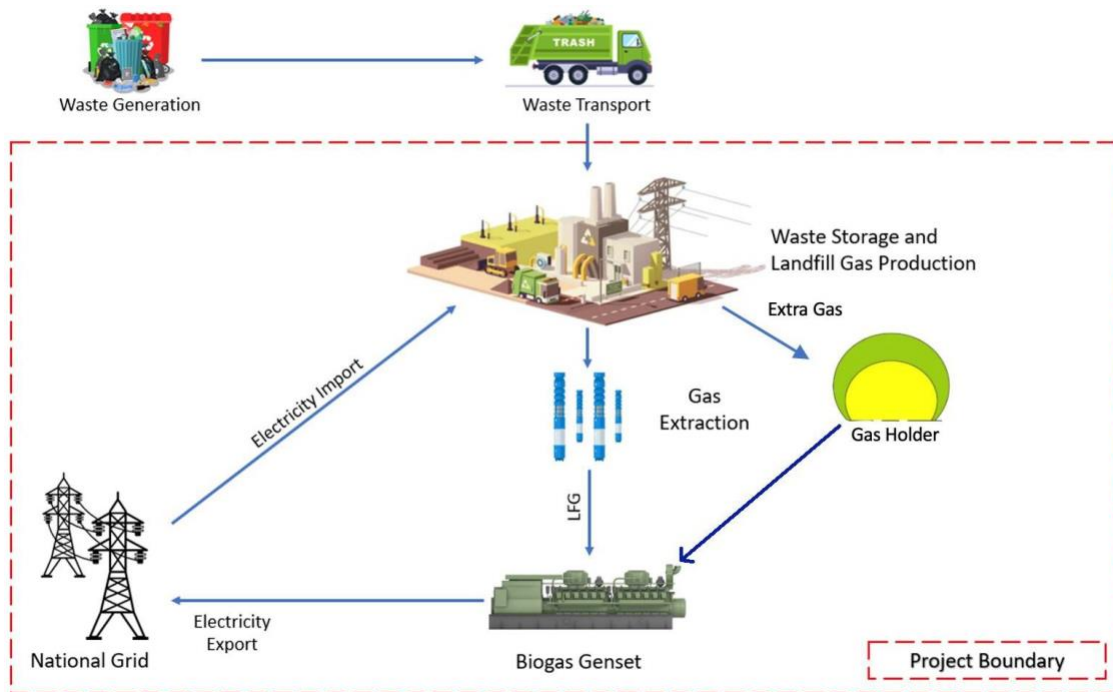


Figure 3 Boundary and the process flow of the project

The collection system which consists of pipelines and booster systems extracts gas from landfill and transmits LFG to chiller unit.

Chiller unit treats LFG and removes impurities and moisture from the gas. LFG is then transmitted to gensets.



Figure 4 Ürker Chiller UC ash 15.1⁵

Table 3 Ciller technical specifications

Brand	Ürker Chillers
Model	UC ASH 15.1
Compressor Type	Semi-hermetic
Production year	2021
Chilling capacity	37.24 kW/h
Total power drawn	13.58 kW/h

The gas holder is designed to securely store excess LFG gas using durable, corrosion - resistant materials and reliable sealing mechanisms. It features multiple layers, including an inner membrane and an outer protective layer, ensuring safety and integrity. The holder's capacity is based on LFG gas production and consumption rates, preventing overpressure and hazards during genset shutdowns. After the LFG is retained in the gas holder, it is utilized to produce electricity once the genset is operational again. This system ensures a continuous LFG gas supply, accommodating production fluctuations without leaks or emissions.

Table 4 Gas holder technical specifications

Brand	Ecomembrane
Model	2MASTER®
External Shape	$\frac{3}{4}$ sphere
Production year	2021
Volume	3500 m ³

⁵ <https://www.urkerchillers.com/tr/img/pdf/Hava-Sogutmalı-Chiller-Katalogu.pdf>

The Project will utilize 2 units of Jenbacher brand gensets with a capacity of 1.414 MWe each. Currently only one unit is in operation and the second unit is planned to start operation in 2025. The Project will have the capacity to generate 2.828 MWe when it reaches full capacity.

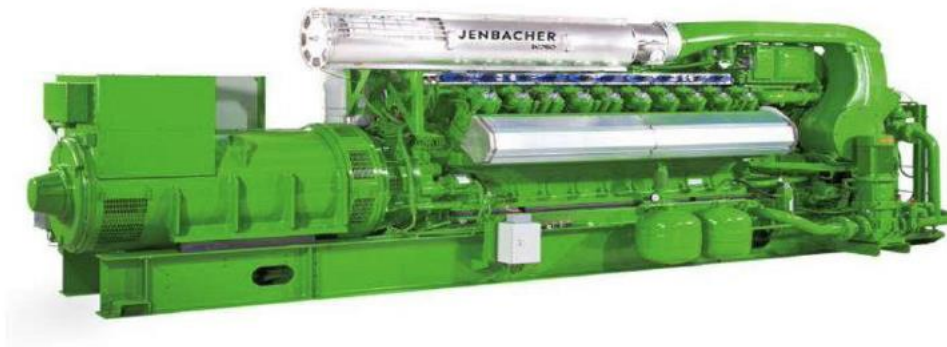
Technical Description

Genset

JGS 420 GS-L.L

static Grid Code

Sinop LF-2020-1



Electrical output 1414 kW el.

Emission values

NOx < 500 mg/Nm³ (5% O₂) | < 190 mg/Nm³ (15% O₂)

Figure 5 Genset technical specifications

Table 5 Genset technical specifications

0.01 Technical Data (at genset)

			100%	75%	50%
Power input	[2]	kW	3.403	2.625	1.848
Gas volume	*)	Nm³/h	851	656	462
Mechanical output	[1]	kW	1.453	1.089	727
Electrical output	[4]	kW el.	1.414	1.061	706
Heat to be dissipated (calculated with Glykol 37%)					
~ Intercooler 1st stage (Engine jacket water cooling circuit)	[9]	kW	291		
~ Intercooler 2nd stage (Low temperature circuit)		kW	75		
~ Lube oil (Engine jacket water cooling circuit)		kW	201		
~ Jacket water		kW	383		
~ Surface heat	ca. [7]	kW	122		
Spec. fuel consumption of engine electric	[2]	kWh/kWel.h	2,41	2,48	2,62
Spec. fuel consumption of engine	[2]	kWh/kWh	2,34	2,41	2,54
Lube oil consumption	ca. [3]	kg/h	0,29	~	~
Electrical efficiency	up to 35°C T1		41,5%	40,4%	38,2%

During the monitoring period gross power generation is 9,183.85 MWh and emission reduction is 43,752 tCO₂e.

The Sinop Landfill Provisional Acceptance the start date of the project is 29-June-2021. However, the project started to utilize methane and generate electricity as of 01-July-2021 which can be seen in EPIAS records. Therefore, start date of the project is 01-July-2021 as per VCS standard 4.4 paragraph 3.8. During the operation phase of the project there is no major event that impacts the GHG emission reductions occurred. Milestones of the project are given in the table below:

Table 6 Milestones of the project activity

Date	Action
December 2020	Feasibility Report
11-December-2020	Operation Contract Between Project Owner and Sinop Coastal Municipalities Union
16-June-2021	EIA Exemption Letter
24-June-2021	EMRA Generation License
25-June-2021	Grid Connection Agreement
29-June- 2021	Sinop Landfill Provisional Acceptance
01-July-2021	Start of Operation and Crediting Period
18-October-2022	Local Stakeholder Consultation Meeting

3.2 Deviations

3.2.1 Methodology Deviations

Not applicable

3.2.2 Project Description Deviations

The carbon consultant from the previous audit period of the project has changed.

Currently, only one unit gas engine is in operation and the second unit is planned to start operation in 2025.

3.3 Grouped Projects

Not applicable

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes ☒ No

Baseline scenario has not changed during the monitoring period.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	OX
Data unit	Dimensionless
Description	Fraction of methane that would be oxidized in the top layer of the SWDS in the baseline
Source of data	Consistent with how oxidation is accounted for in the methodological tool "Emissions from solid waste disposal sites"
Value applied	0.1

Justification of choice of data or description of measurement methods and procedures applied	Default Value as per AMS-III.G v10
Purpose of Data	Baseline emission calculation
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC AR5 ²²
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	Default value of 28 from IPCC fifth Assessment Report (AR5). Shall be updated according to any future COP/MOP decisions.
Purpose of Data	Ex ante calculation of baseline emissions
Comments	-

Data / Parameter	η_{PJ}
Data unit	%
Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	Feasibility study
Value applied	85%
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of data	Ex Ante Baseline emission calculation
Comments	N/A

Data / Parameter	φ_y
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Data unit	-
Description	Default value for the model correction factor to account for model uncertainties
Source of data	<i>"Methodological Tool: Emissions from solid waste disposal sites" (Version 08.0)</i>
Value applied	0.75
Justification of choice of data or description of measurement methods and procedures applied	The project selects Option 1 :Use a default value to determine ϕ y in the tool "Emissions from solid waste disposal sites" (version 08.0), and the project belongs to the situation of Application A: The project activity mitigates methane emissions from a specific existing SWDS, thus 0.75 is chosen for ϕ y
Purpose of data	Ex Ante baseline emission calculation
Comments	N/A

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Default Value as per Tool04 - "Emissions from solid waste disposal sites"
Purpose of data	Ex Ante baseline emission calculation
Comments	N/A

Data / Parameter	$DOC_{f,y}$
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Default Value as per Tool04 - "Emissions from solid waste disposal sites"

Purpose of data	Ex Ante baseline emission calculation
Comments	N/A

Data / Parameter	MCF _y
Data unit	-
Description	Methane correction factor
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	1
Justification of choice of data or description of measurement methods and procedures applied	The landfill is controlled, including the following activities: (i) cover material; (ii) mechanical compacting; (iii) levelling of the waste. Therefore, 1.0 for anaerobic managed solid waste disposal sites is recommended by "Emissions from solid waste disposal sites (Version 08.0)."
Purpose of data	Baseline emission calculation
Comments	N/A

Data / Parameter	W _x																
Data unit	tons																
Description	Total amount of waste disposed in a SWDS in the year x																
Source of data	Feasibility study report (FSR) has been used for the waste amount. Waste compositions study of incoming waste to the landfill has been used for waste composition.																
Value applied	<u>Waste Amount:</u> <table border="1"> <thead> <tr> <th>Year</th><th>Total (tonnes)</th></tr> </thead> <tbody> <tr> <td>2012</td><td>18,250</td></tr> <tr> <td>2013</td><td>27,375</td></tr> <tr> <td>2014</td><td>28,744</td></tr> <tr> <td>2015</td><td>30,181</td></tr> <tr> <td>2016</td><td>31,690</td></tr> <tr> <td>2017</td><td>33,274</td></tr> <tr> <td>2018</td><td>36,269</td></tr> </tbody> </table>	Year	Total (tonnes)	2012	18,250	2013	27,375	2014	28,744	2015	30,181	2016	31,690	2017	33,274	2018	36,269
Year	Total (tonnes)																
2012	18,250																
2013	27,375																
2014	28,744																
2015	30,181																
2016	31,690																
2017	33,274																
2018	36,269																

Waste Type	Weight Fraction (%)
Food	58.59
Paper and Cardboard	7.06
Wood	5.90
Textile	3.66
Garden, yard and park waste	0.00
Inert materials (plastic, metal etc.)	24.79
Total	100.00

2019	38,083
2020	43,414
2021	56,438
2022	70,548
2023	71,959
2024	73,398
2025	74,132
2026	74,874
2027	75,622
2028	76,379
2029	77,142
2030	77,914

During the monitoring period;

Year	Ton
01-March-2023 to 31-December-2023	44,627.96
01-January-2024 to 31-August-2024	34,103.14
Total	78,731.11

	<u>Waste Composition:</u>
Justification of choice of data or description of measurement methods and procedures applied	It is calculated as total waste amount dumped in the landfill site in the year x from FSR. For waste composition, sample amount of waste from different areas classed according to their wealth is taken and studied to determine waste composition by weight.
Purpose of data	Ex-ante calculation of baseline emissions
Comments	N/A

Data / Parameter	DOC _j														
Data unit	-														
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)														
Source of data	<i>IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)</i>														
Value applied	<table> <tr> <th>Waste type j</th><th>DOC_j (%wet waste)</th></tr> <tr> <td>Wood and wood products</td><td>43</td></tr> <tr> <td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr> <tr> <td>Food, food waste, beverages and tobacco (other than sludge)</td><td>15</td></tr> <tr> <td>Textiles</td><td>24</td></tr> <tr> <td>Garden, yard and park waste</td><td>20</td></tr> <tr> <td>Glass, plastic, metal other inert waste</td><td>0</td></tr> </table>	Waste type j	DOC _j (%wet waste)	Wood and wood products	43	Pulp, paper and cardboard (other than sludge)	40	Food, food waste, beverages and tobacco (other than sludge)	15	Textiles	24	Garden, yard and park waste	20	Glass, plastic, metal other inert waste	0
Waste type j	DOC _j (%wet waste)														
Wood and wood products	43														
Pulp, paper and cardboard (other than sludge)	40														
Food, food waste, beverages and tobacco (other than sludge)	15														
Textiles	24														
Garden, yard and park waste	20														
Glass, plastic, metal other inert waste	0														
Justification of choice of data or description of measurement methods and procedures applied	Application A in the Tool04 has been chosen.														

Purpose of data	Ex ante baseline emission calculation
Comments	N/A

Data / Parameter	kj					
Data unit	1/yr					
Description	Decay rate for the waste type j					
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 3.3)					
Value applied	Waste Tpe j		Boreal and temperature (MAT < 20°C)		Tropical (MAT>20°C)	
			Dry (MAP/P ET < 1)	Wet (MAP/P ET >1)	Dry(MAP<1 000 mm)	Wet (MAP>10 00 mm)
	Slowly degrading	Pulp, paper and cardboard (other than sludge)	0.04	0.06	0.045	0.07
		Wood, wood products and straw	0.02	0.03	0.025	0.035
	Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.05	0.10	0.065	0.17
	Rapidly degrading	Food, food waste, sewage, sludge,bever ages and tobacco	0.06	0.185	0.085	0.40
Justification of choice of data or description of measurement methods and procedures applied	Climatic conditions for Sinop where project site located in: MAT – mean annual temperature: 14.3 °C MAP – Mean annual precipitation: 689.3 mm PET – potential evapotranspiration: 150 mm					

	Therefore, Boreal ($\text{MAT} \leq 20^{\circ}\text{C}$) and wet ($\text{MAP/PET} > 1$) conditions are observed in the project site. <i>Sources:</i> https://www.mgm.gov.tr/veridegerlendirme/il-ve-ilceleri-istatistik.aspx?k=undefined&m=SINOP https://www.mgm.gov.tr/arastirma/buharlasma.aspx?s=2020
Purpose of data	Ex ante baseline emission calculation
Comments	N/A

Data / Parameter	ρ_{CH_4}
Data unit	$\text{t CH}_4 / \text{m}^3 \text{CH}_4$
Description	Density of methane at normal conditions
Source of data	‘Thermophysical properties of fluids. II. Methane, Ethane, Propane, Isobutane and Normal Butane’ by B.A. Younglove, J.F. Ely Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Value applied	0.00067
Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Purpose of data	Calculation of baseline/project emission calculation
Comments	Corrected to reference conditions

Data / Parameter	NCVCH ₄
Data unit	MJ/Nm^3
Description	Net calorific value of methane at reference conditions
Source of data	AMS-III.G. “Landfill methane recovery” (version 10.0).
Value applied	35.9
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of data	Calculation of baseline emission
Comments	N/A

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y
Source of data	Calculated by Turkish Ministry of Energy and Natural Resources. ⁶ Also, "Tool to calculate the emission factor for an electricity system (Version 07.0)" is applied.
Value applied	0.5552
Justification of choice of data or description of measurement methods and procedures applied	The data was obtained from Turkish Ministry of Energy and Natural Resources. Also, "Tool to calculate the emission factor for an electricity system (Version 07.0)" is applied.
Purpose of data	Calculation of baseline emission
Comments	N/A

Data / Parameter	TDLy
Data unit	%
Description	Average technical transmission and distribution losses
Source of data	Obtained from https://webapi.teias.gov.tr/file/9248a331-e529-47ca-8c49-22f8705f5810?download
Value applied	10.0%
Justification of choice of data or description of measurement methods and procedures applied	Most recent annual data is available for 2020 and losses from transmission and distribution are 1.7% and 8.3% respectively and total of 10%.
Purpose of data	Calculation of project emission
Comments	N/A

4.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}
Data unit	MWh
Description	Quantity of net electricity supplied by the project activity to the grid in year y

⁶ <https://enerji.gov.tr/evced-cevre-ve-iklim-turkiye-ulusal-elektrik-sebekesi-emisyon-faktoru>

Source of data	Electricity meter															
Description of measurement methods and procedures to be applied	The quantity of electricity consumed by the project plant is monitored based on the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate.															
Frequency of monitoring/recording	Continuously measured and monthly recorded.															
Value monitored	8,788.376 <table><tr><td>Year</td><td>Net Electricity supplied to the grid(MWh)</td></tr><tr><td>01-March-2023 to 31-December-2023</td><td>4,742.92</td></tr><tr><td>01-January-2024 to 31-August-2024</td><td>4,045.45</td></tr></table>	Year	Net Electricity supplied to the grid(MWh)	01-March-2023 to 31-December-2023	4,742.92	01-January-2024 to 31-August-2024	4,045.45									
Year	Net Electricity supplied to the grid(MWh)															
01-March-2023 to 31-December-2023	4,742.92															
01-January-2024 to 31-August-2024	4,045.45															
Monitoring equipment	The quantity of electricity consumed by the project plant is monitored based on the main meter (also referred as TEİAŞ meters) readings and/or through protocols where and when appropriate. Electricity meters; <table><tr><td>Equipment</td><td>Serial Number</td><td>Brand Model</td><td>Accuracy Class</td><td>Date of Calibration</td></tr><tr><td>Main meter</td><td>80341274</td><td>Makel-C520KMT2556</td><td>C Class (0.5)</td><td>01-July-2022</td></tr><tr><td>Backup meter</td><td>80272124</td><td>Makel-CC20KMT2556</td><td>C Class (0.5)</td><td>26-May-2021</td></tr></table>	Equipment	Serial Number	Brand Model	Accuracy Class	Date of Calibration	Main meter	80341274	Makel-C520KMT2556	C Class (0.5)	01-July-2022	Backup meter	80272124	Makel-CC20KMT2556	C Class (0.5)	26-May-2021
Equipment	Serial Number	Brand Model	Accuracy Class	Date of Calibration												
Main meter	80341274	Makel-C520KMT2556	C Class (0.5)	01-July-2022												
Backup meter	80272124	Makel-CC20KMT2556	C Class (0.5)	26-May-2021												
QA/QC procedures to be applied	The electricity meters are used to measure the net electricity exported to the grid and imported from the grid and used for billing purposes. Therefore, the meters are sealed by TEİAŞ (Türkiye Electricity Transmission CO) and/or through protocols where and when appropriate. The grid company is responsible for maintenance and calibration of the device. In accordance with the “Regulation on Measurement and Measuring Tools”, electricity meters have to be calibrated within a frequency of once per 10 years.															
Purpose of the data	Calculation of baseline emission															

Calculation method	This parameter is calculated by subtracting electricity imported from electricity supplied to the grid and formula is as follows: $EG_{PJ,y} = EG_{\text{export},y} - EG_{\text{import},y}$
Comments	The data is monitored by the bi-directional electricity meter that installed at the outlet of the onsite substation with the accuracy of 0.5. Meter readings are read and recorded by YEDAS (Distribution Company) and EPIAS. Since the electricity is transferred via energy transmission line EPIAS deduct this amount and EPIAS system shows invoice-based reading by excluding transmission losses. EPIAS data is the main data and YEDAS readings (OSOS records) used for cross-check. Turkish legislation enforces calibration on electricity meters every 10 years⁷ Due to a communication failure, the electricity meter in the field was replaced by YEDAŞ mobile teams on 01-July-2022.⁸

Data / Parameter	FCH _{4,BL,R,y}
Data unit	tCH ₄ /yr
Description	Amount of methane in the LFG which is flared due to a requirement in year y
Source of data	Regulation in the country
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	N/A
Value monitored	0
Monitoring equipment	
QA/QC procedures to be applied	N/A

⁷ <https://www.resmigazete.gov.tr/arsiv/22000.pdf>

⁸ Meter change document

Purpose of the data	N/A
Calculation method	N/A
Comments	Since there is no regulation Case 1 is valid during monitoring period.

Data / Parameter	$p_{reg,y}$
Data unit	Dimesionless
Description	Fraction of LFG that is required to be flared due to a requirement in year y
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odour concerns
Description of measurement methods and procedures to be applied	As per regulation of the country there is no obligation to utilize or destroy methane.
Frequency of monitoring/recording	N/A
Value monitored	0
Monitoring equipment	
QA/QC procedures to be applied	N/A
Purpose of the data	N/A
Calculation method	N/A
Comments	Since there is no regulation Case 1 is valid during monitoring period.

Data / Parameter	EG_y
Data unit	MWh
Description	Quantity of gross electricity generation
Source of data	SCADA

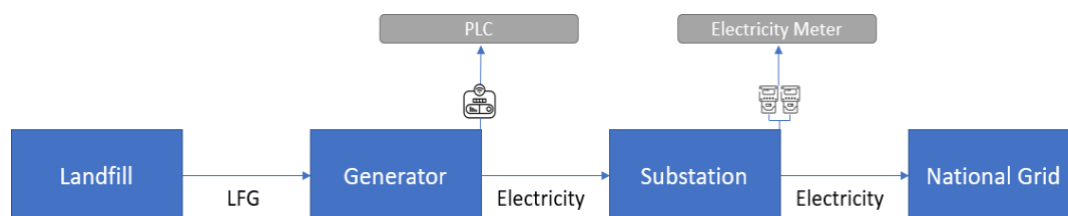
Description of measurement methods and procedures to be applied	The quantity of generated electricity monitored by SCADA meter on generator with PLC system. Measurements are monitored continuously and recorded by project staff monthly basis. Quantity of electricity generation is monitored by PLC. PLC specifications are shown below; <table><tr><td>Equipment</td><td>Brand&Model</td><td>Rated Voltage</td><td>Supply Voltage</td></tr><tr><td>PLC</td><td>Bender-FP200</td><td>50 V</td><td>24 V</td></tr></table>			Equipment	Brand&Model	Rated Voltage	Supply Voltage	PLC	Bender-FP200	50 V	24 V
Equipment	Brand&Model	Rated Voltage	Supply Voltage								
PLC	Bender-FP200	50 V	24 V								
Frequency of monitoring/recording	Continuously										
Value monitored	9,183.85 <table><tr><td>Year</td><td>Electricity Generated MWh)</td></tr><tr><td>01-March-2023 to 31-December-2023</td><td>4,956.36</td></tr><tr><td>01-January-2024 to 31-August-2024</td><td>4,227.50</td></tr></table>			Year	Electricity Generated MWh)	01-March-2023 to 31-December-2023	4,956.36	01-January-2024 to 31-August-2024	4,227.50		
Year	Electricity Generated MWh)										
01-March-2023 to 31-December-2023	4,956.36										
01-January-2024 to 31-August-2024	4,227.50										
Monitoring equipment	PLC										
QA/QC procedures to be applied	N/A										
Purpose of the data	Calculation of emission reduction										
Calculation method	N/A										
Comments	Since emission reduction calculated from electricity generation, this parameter shall be used.										

Data / Parameter	EE _y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	AMS-III.G "Landfill methane recovery" (version 10.0)
Description of measurement methods and procedures to be applied	Default value in AMS-III.G v 10 has been used.

Frequency of monitoring/recording	N/A
Value monitored	40
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of the data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

4.3 Monitoring Plan

Monitoring follows the procedures set by applicable methodology and all applicable tools of the above explained parameters. The monitoring consists of: Responsibilities for monitoring The project owner is responsible for the operation and maintenance of the landfill and the installed equipment. The project owner is also responsible for the administration of the data, setting up a VER team who are responsible for monitoring all data required to estimate emission reductions. Data and parameters to be monitored:



Monitoring Device	Monitored Parameter	Description
PLC	EG _y	Amount of gross electricity generated on alternator of genset using LFG by the project activity is continuously monitored by PLC device installed on alternator of the generator and continuously monitored in SCADA.
Electricity Meter	EG _{PJ,y}	Quantity of net electricity generation supplied by the project to the grid is

		continuously monitored by bi-directional electricity meter installed by outlet of onsite substation.
--	--	--

All data collected as part of monitoring is being archived electronically by the Project Proponent and be kept at least for 2 years after the end of the last crediting period.

Registration of the monitored data

The Sinop LFG VER-Team are responsible for quality assurance and quality control of the monitoring equipment. The data measured by the monitoring equipment is stored and processed into a monitoring report, which is submitted by the project owner. All the monitored data will be stored during the crediting period and for at least two years after the end of the crediting period, whichever occurs later.

Corrective actions and emergency preparedness

The VER-Team regularly check the monitoring system for errors. In the case of errors, corrective actions are undertaken by the VER-Team, or if required, by the supplier of the monitoring equipment.

QA/QC procedure

Strong quality assurance and quality control procedures are in place to monitor the equipment and data collection. Equipment and facilities are subject to a regular maintenance and testing regime to ensure accuracy following the supplier's manual. In case of data deficiency or incorrect data reading, the data is crosschecked with other parameters and data leading to the most conservative emission reductions are considered for the calculation of the emission reductions.

Measuring of exported and imported electricity

The main meter that monitors electricity import and export is placed at the grid connection. Monthly protocols based on the readings that are confirmed by both TEİAŞ and the project participant are prepared by the end of each month and stored. These protocols are also the basis of invoicing for the electricity sales from the proposed project activity. The electricity meters are bidirectional and measure both the imported and exported electricity from and to the grid.

Organization chart

Organization chart of the facility is shown below;

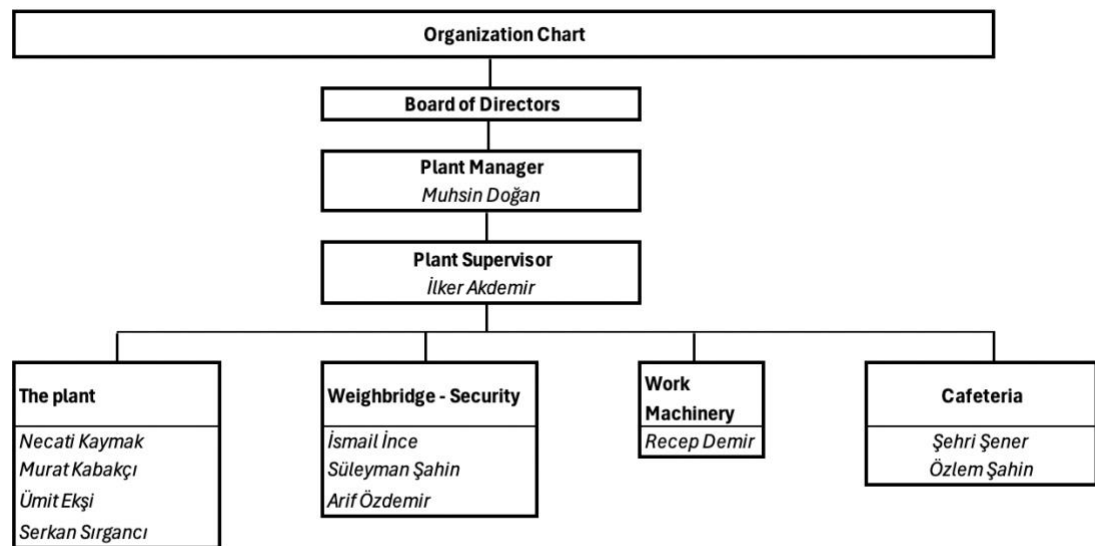


Figure 6 Organization Chart

Leakage Management

The plant's leachate is being sent to the Gerze Municipality Wastewater Treatment Plant.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emission reductions are calculated in accordance with AMS-III.G (Version 10.0) and AMS-I.D. (Version 18.0). The baseline emissions are calculated by the following equation;

Equation

$$BE_y = BE_{CH_4,y} + BE_{\text{electricity},y}$$

Where;

BE_y : Baseline emissions in year y (tCO₂e/y)

$BE_{CH_4,y}$: Baseline emissions of methane from the SWDS in year y (tCO₂e/y)

$BE_{\text{electricity},y}$: Baseline emissions associated with electricity displacement y (tCO₂e/y)

Determination of Baseline Emissions from AMS-III.G. ($BE_{CH_4,y}$)

As per AMS-III.G, the actual emission reduction achieved for Type III component of the project (which means not including emission reduction from Type I component, i.e. LFG power generation) during the crediting period calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

Equation

$$ER_{y,calculated} = (1 - OX) \times (F_{CH_4,PJ,y} - F_{CH_4,BL,y}) \times GWP_{CH_4} - PE_y - LE_y$$

Where:

$F_{CH_4,PJ,y}$: Methane captured and destroyed/gainfully used by the project activity in the year y (tCH₄)

Determination of $F_{CH_4,BL,y}$,

As per ACM0001 v19.0, 4 cases given in the table below has been evaluated

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Since there is no existing LFG capture system prior to project activity and there is no requirement to destroy methane Case 1 is the project activity case. Therefore, $F_{CH_4,BL,y}$ equals to 0.

Equation

$$F_{CH_4,PJ,y} = D_{CH_4,y} \times w_{CH_4,y} \times \sum_i LFG_{i,y}$$

Where:

- $LFG_{i,y}$ = Landfill gas destroyed via method i (flaring, fuelling, combustion, injection to a grid, etc.) in year y (m³LFG). The flow or volume measurement shall be made either on a dry basis or at the same humidity as $w_{CH_4,y}$
- $w_{CH_4,y}$ = Methane content in landfill gas in year y (volume fraction, m³CH₄/m³LFG). Landfill gas composition shall be measured either on a dry basis or at the same humidity as used to determine $LFG_{i,y}$

D_{CH4} = Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/m³). If $LFG_{i,y}$ is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions

For project activities that utilize the recovered methane for power generation, $F_{CH4,PJ,y}$ may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

Equation

$$F_{CH4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4}$$

Where:

- EG_y = Electricity generation in year y (MWh)
- 3600 = Conversion factor (1 MWh = 3600 MJ)
- NCV_{CH4} = NCV of methane (MJ/Nm³) use default value: 35.9 MJ/Nm³
- EE_y = Energy Conversion Efficiency of the project equipment determined from one of the following options:
 - Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
 - Default efficiency of 40 per cent

There is no mandatory regulatory requirement on landfill gas destruction/combustion that has been systematically enforced in Turkey which means there is no contractual requirement on methane destruction that the project owner is bound to. Therefore, $F_{CH4,BL,y}$ is 0.

According to equations above EG_y , $F_{CH4,PJ,y}$ and finally $BE_{CH4,y}$ in this monitoring is given in the table below:

Months	EG_y (MWh)	$F_{CH4,PJ,y}$ (tCH ₄)	$F_{CH4,BL,y}$ (tCH ₄)	BE_{CH4} (tCO _{2e})
March 2023	481.34	80.85	0	2,037.41

April 2023	498.14	83.67	0	2,108.49
May 2023	483.32	81.18	0	2,045.77
June 2023	546.07	91.72	0	2,311.39
July 2023	538.50	90.45	0	2,279.36
August 2023	432.51	72.65	0	1,830.73
September 2023	427.71	71.84	0	1,810.38
October 2023	478.48	80.37	0	2,025.28
November 2023	490.26	82.35	0	2,075.15
December 2023	580.02	97.42	0	2,455.10
January 2024	608.38	102.19	0	2,575.12
February 2024	537.36	90.26	0	2,274.50
March 2024	531.06	89.20	0	2,247.87
April 2024	502.50	84.40	0	2,126.95
May 2024	545.51	91.63	0	2,309.03
June 2024	534.04	89.70	0	2,260.45
July 2024	430.58	72.32	0	1,822.53
August 2024	538.07	90.38	0	2,277.54
Total Sum	9,183	1,542	0	38,873

Determination of Baseline Emissions from AMS-III.G. ($BE_{\text{electricity},y}$)

Project activity utilize LFG to generate renewable energy and generated electricity is supplied to the national grid.

As per AMS-I.D. (version 18.0) baseline emission calculated as follows:

Equation

$$BE_{y, \text{electricity}} = EG_{PJ,y} \times EF_{\text{grid},y}$$

Where;

- BE_{y,electricity} = Baseline emissions in year t (t CO₂)
- EG_{PJ,facility,y} = Quantity of net electricity generation that is produced by the project plant to the grid in year y (MWh)
- EF_{grid,y} = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

Emission factors of the Turkish National Grid is as follows by the publications of Ministry of Energy and Natural Sources⁹

- Operating Margin-OM; 0.7424 tCO₂/MWh
- Build Margin-BM; 0.3680 tCO₂/MWh

Combined Margin-CM (for other renewable than solar and wind); 0.5552 tCO₂/MWh

Therefore, baseline emissions related to electricity generations is as follows;

Months	EG _{PJ,facility,y} (Net Electricity, Mwh)	BE _{electricity,y} (tCO _{2e})
March 2023	460.62	255.73
April 2023	476.68	264.66
May 2023	462.51	256.78
June 2023	522.56	290.12
July 2023	515.31	286.10
August 2023	413.89	229.79
September 2023	409.29	227.24
October 2023	457.87	254,21
November 2023	469.15	260,47
December 2023	555.05	308,16
January 2024	582.18	323,23
February 2024	514.22	285,49
March 2024	508.20	282,15
April 2024	480.86	266,97
May 2024	522.02	289,83
June 2024	511.04	283,73
July 2024	412.04	228,76
August 2024	514.90	285,87
Total Sum	8,788.376	4,879

As a result of these, sum of baseline emission BE_y is as follows:

Months	BE _{electricity,y} (tCO _{2e})	BE _{CH₄} (tCO _{2e})	BE _y (tCO _{2e})
March 2023	255.73	2,037.41	2,293.14

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https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUES EmisyonFktr/Belgeler/TUESEF_Bilgi_Formu.pdf

April 2023	264.66	2,108.49	2,373.14
May 2023	256.78	2,045.77	2,302.55
June 2023	290.12	2,311.39	2,601.52
July 2023	286.10	2,279.36	2,565.46
August 2023	229.79	1,830.73	2,060.52
September 2023	227.24	1,810.38	2,037.61
October 2023	254.21	2,025.28	2,279.49
November 2023	260.47	2,075.15	2,335.62
December 2023	308.16	2,455.10	2,763.26
January 2024	323.23	2,575.12	2,898.34
February 2024	285.49	2,274.50	2,559.99
March 2024	282.15	2,247.87	2,530.02
April 2024	266.97	2,126.95	2,393.92
May 2024	289.83	2,309.03	2,598.86
June 2024	283.73	2,260.45	2,544.18
July 2024	228.76	1,822.53	2,051.29
August 2024	285.87	2,277.54	2,563.41
Total Sum	4,879	38,873	43,752

5.2 Project Emissions

Project Emissions as per AMS-III.G.

Project emissions are calculated as follows as per AMS-III.G v10.

Equation

$$PE_y = PE_{Power,y} + PE_{flare,y} + PE_{process,y}$$

Where:

PE_y	=	Project emissions in year y (t CO ₂ /yr)
$PE_{Power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO ₂ e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the landfill gas stream in the year y (t CO ₂ e)
$PE_{process,y}$	=	Emissions from the landfill gas upgrading process in the year y (t CO ₂ e), determined by following the relevant procedures described in annex 1 of AMS-III.H.

Since the project activity involves electricity generation, it is that the plant has generated more electricity than it consumes. Moreover, net electricity generation and emission reduction due to renewable energy generation are claimed under AMS-I.D. v18.0 and energy consumption is subtracted from energy generated and EPIAS data has already shows transmission and distribution losses for the electricity generated to grid and consumed from the grid. Therefore, PE_{power} is zero.

In the project site, there is gas holder to retain extra gas, there is no flare unit. Hence PE_{flare} is taken as zero.

The project does not involve landfill gas upgrading process, hence $PE_{process}$ is also zero.

Project Emissions as per AMS-I.D.

As per the paragraph 39 of AMS-I.D., version 18.0., for most renewable energy project activities, $PE_y = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in ACM0002 v21.0:

- (a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);
- (b) Emissions from water reservoirs of hydro power plants.

As per the paragraph 40 of AMS-I.D., CO₂ emissions from on-site consumption of fossil fuels due to the project activity shall be calculated using the latest version of the “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”. Also, the paragraph 41 of AMS-I.D. states that in case biomass is sourced from dedicated plantations, the procedures in the tool “Project emissions from cultivation of biomass” shall be used

Since none of the exception stated in the relevant paragraphs of AMS-I.D. mentioned above is applicable to the proposed project activity, PE_y as per AMS-I.D., version 18.0., shall be accounted as 0.

In conclusion, PE_y for the project activity is 0.

5.3 Leakage Emissions

No leakage effects are accounted for under AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0).

5.4 GHG Emission Reductions and Carbon Dioxide Removals

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
01-March-2023 to 31-December-2023	23,612	0	0	23,612	0	23,612
01-January-2024 to 31-August-2024	20,140	0	0	20,140	0	20,140
Total	43,752			43,752		43,752

State the non-permanence risk rating (%)	Not applicable
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☒ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	N/A
Has the LTA been updated based on monitored data, if applicable?	N/A
State, in tCO ₂ e, the expected total GHG benefit to date.	N/A
If a loss occurred (including a loss event or reversal), state the amount of tCO ₂ e lost:	N/A

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
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01-March-2023 to 31-December-2023	27,821.69	23,612	-17.83%	Ex-ante calculations are higher than ex-post emission reductions. For ex-ante emissions, there are many assumptions to calculate emission reductions, so the difference is acceptable such project.
01-January-2024 to 31-August-2024	28,192.36	20,140	-39.98%	Ex-ante calculations are higher than ex-post emission reductions. During the ex-ante calculations, it was planned to commission the second gas engine in 2024. However, since the second gas engine was not commissioned in 2024, it is acceptable for the difference between the actual and estimated values to be high.
Total	56,014.05	43,752	-28.03%	Ex-ante calculations are higher than ex-post emission reductions. For ex-ante emissions, there are many assumptions to calculate emission reductions, so the difference is acceptable such project.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

N/A